

Evaluation of Surface and Bulk Properties of Alkali Halides: A First-Principles Study on (100) and (110) Facets

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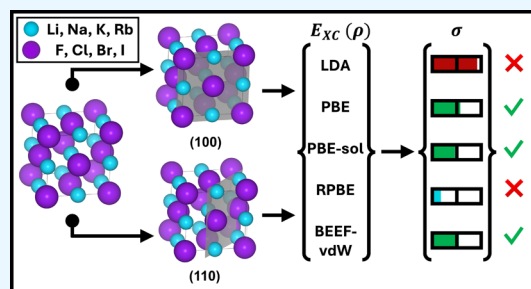


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ABSTRACT: Five exchange-correlation functionals (LDA, PBE, RPBE, PBE-sol, and BEEF-vdW) are studied for their accuracy in predicting bulk and surface properties of 16 alkali halides. Lattice structures, formation energies, and the surface energies of (100) and (110) facets are calculated and compared against experimental values and classical electrostatic models. While all functionals capture broad trends across the halide series, notable differences emerge in both quantitative and qualitative reliability. PBE-sol and BEEF-vdW deliver the most balanced performance across all metrics, whereas LDA and RPBE are poor performers, with larger deviations, due to over- and under-binding tendencies. Surface energy anisotropy is shown to depend on the choice of functional. The results highlight the importance of selecting appropriate methods not only for quantitative predictions but also for preserving physically meaningful descriptions of materials and surfaces. This work provides a practical benchmark for modeling ionic solids and surfaces.



1. INTRODUCTION

Alkali halides (AX, where A^+ is an alkali metal, and X^- is a halide ion) are ionic crystals that have historically served as benchmark systems in solid-state physics and material chemistry.^{1–8} Their highly ionic bonding and simple cubic structure make them ideal for investigating relationships between crystal structure, bonding, and surface energetics. Furthermore, both cations and anions follow clear, systematic trends, allowing for simple interpretation of observed results of the materials which they form.^{9–11} Accurate prediction of their bulk properties, such as lattice parameters and formation energies, as well as surface properties, including surface energies and anisotropy, is crucial for fundamental understanding of the involved physics and subsequent applications in crystal growth,^{12,13} medical imaging,¹⁴ and ion transport across materials and interfaces.^{15,16}

Modeling the thermodynamic and structural properties of alkali halides has a long history, beginning with classical approaches based on electrostatics and interatomic potentials. The pioneering works of Born and Huang¹ and Fumi and Tosi^{17–19,10,20,21} established foundational models for lattice and ionic bonding descriptions, which were later extended to other phenomena, including surface models.^{22–30} Building on these developments, Tasker introduced a classification of surfaces based on polarity and used classical electrostatics to compute ideal surface energies for low-index facets.⁹ Although these classical models neglect quantum effects such as charge redistribution and polarization at surfaces, their simplicity and physical transparency have made them valuable reference points for understanding trends in ionic crystal behavior.

Density Functional Theory (DFT) has later enabled the computation of bulk and surface properties of these materials

with greater accuracy. DFT incorporates the quantum mechanical nature of electrons, which enables the prediction of optimized lattice geometries and surface energetics without the need for empirical fitting. Several prior studies have applied DFT to specific alkali halides to investigate electronic structure, lattice, and surface relaxations.^{31–34} The choice of exchange-correlation functional plays a critical role in DFT predictions.^{35,36} Each functional represents an approximation to the exact exchange-correlation energy, varying in their treatment of gradient corrections, dispersion, and electronic localization, which results in significant differences in predicted energies. In this work, we present a comprehensive benchmark of five widely used exchange-correlation functionals in predicting the bulk and surface properties of alkali halides. The functionals are LDA,³⁷ PBE,³⁸ RPBE,³⁹ PBE-sol,⁴⁰ and BEEF-vdW.⁴¹

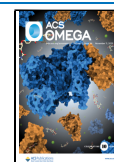
We evaluate each functional's performance in predicting lattice constants, enthalpies of formation, and surface energies for both the (100) and (110) surface facets. Although the polar (111) facets have attracted research interest for their unique properties,^{42,43} our benchmark is focused on the fundamental nonpolar (100) and (110) facets, as they constitute the lowest-energy cleavage planes and are thus the most representative for a systematic computational study. Comparisons are made to both

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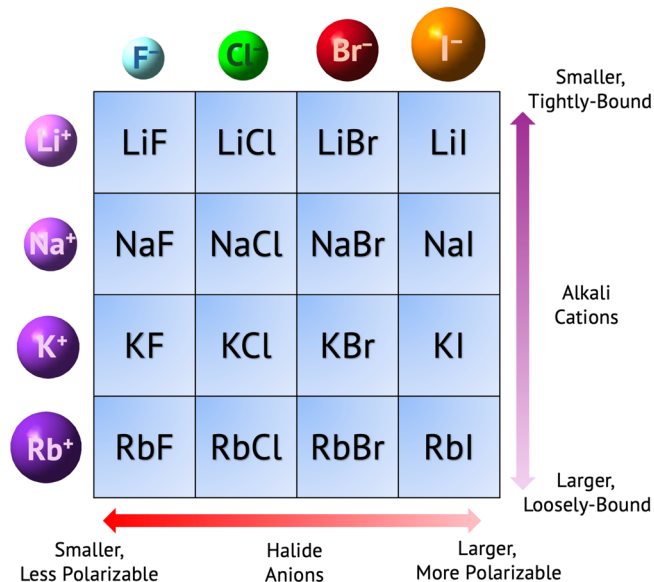
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experimentally available data and Tasker's classical model surface energies.⁹ Reliable experimental surface energies for alkali halide crystals remain scarce, and the limited available data shows significant inconsistencies. As reported by Norman et al.,⁴⁴ surface energy values obtained from different techniques vary widely, and in some cases, differ by several orders of magnitude, even for the same material. For example, experimentally measured surface energies for NaCl show significant variation, with values ranging from 310 to 75,000 erg cm². This variability makes it difficult to establish a consistent experimental reference, which in turn complicates the assessment of which exchange-correlation functional performs best. Without reliable surface benchmarks, we assess qualitative consistency, such as the ranking of surface facet stability and degree of anisotropy, both of which are key factors in determining Wulff constructions and surface terminations. We find that while many functionals yield reasonable predictions due to partial error cancelation, others exhibit asymmetric deviations from observed trends, which is important in evaluating the robustness of a functional for surface modeling.

Scheme 1 shows the complete set of 16 alkali halides examined in this study, encompassing all combinations of

Scheme 1. 16 Alkali Halides Examined in This Study, Formed by Combining Four Alkali Cations (Li⁺, Na⁺, K⁺, and Rb⁺) with Four Halide Anions (F[−], Cl[−], Br[−], I[−])^a



^aThe grid is organized to reflect periodic trends in ionic size, charge density, and polarizability. The ionic size increases whereas the polarizability decreases, from the top left (small, hard ions) to the bottom right (large, soft ions).

monovalent alkali metals (Li, Na, K, Rb) and halide anions (F, Cl, Br, I). This broad chemical space enables a systematic evaluation of theoretical methods across variations in ion size, polarizability, and bonding strength. By systematically comparing the five exchange-correlation functionals across 16 chemically diverse alkali halides and two crystallographic facets, this study offers critical insights into the reliability of DFT-based predictions for both bulk and surface properties. The results are intended to guide the selection of appropriate exchange-correlation functionals in future computational studies of ionic crystals for their various applications in crystal growth,

nanoparticle Wulff constructions, and in predicting catalytic behavior.^{45–47}

2. METHODS

All spin-unpolarized DFT calculations were performed using the Quantum ESPRESSO software package,^{48–50} with structure setup and management handled using the Atomic Simulation Environment (ASE).⁵¹ The Kohn–Sham wave functions were expanded using a plane-wave basis set with 600 and 6000 eV plane-wave and density cutoffs, respectively, and a Fermi–Dirac smearing of 0.1 eV was applied. As introduced earlier, we performed calculations using a range of DFT functionals commonly used in materials modeling and surface chemistry, including LDA,^{52,37} the GGA functionals PBE,³⁸ RPBE,³⁹ PBEsol,⁴⁰ and GGA with van der Waals (vdW) corrections, BEEF-vdW.⁴¹ Core electrons were treated using the projector augmented-wave (PAW) method,^{53,54} and the valence electron configurations used in the pseudopotentials⁵⁵ are summarized in Table S1. Due to the limited availability of functional-specific PAW pseudopotentials, we used PBE-generated pseudopotentials for all considered GGA-level functionals, and the Perdew–Zunger LDA pseudopotentials,⁵² based on the scheme of Troullier and Martins.^{56,57} Periodic boundary conditions were applied in all directions, with a Γ -centered $5 \times 5 \times 5$ Monkhorst–Pack grid used for bulk calculations, and a $5 \times 5 \times 1$ grid for surfaces.⁵⁸ Electronic minimization was considered to have reached self-consistent-field convergence when the total energy difference between successive steps was $\leq 1 \times 10^{-5}$ eV/cell. Ground-state geometries were obtained by relaxing all atomic positions until the forces on each atom were below 0.03 eV/Å.

All alkali halide bulk structures considered in this study were obtained from the Materials Project database⁵⁹ in their conventional cubic form (space group $Fm\bar{3}m$), each containing four formula units ($Z = 4$). We considered 16 AX-type compounds, where A denotes an alkali metal (Li, Na, K, Rb) and X a halide ion (F, Cl, Br, I). Bulk lattice constants were determined via isotropic volume perturbations of the unit cell. The optimized lattice constants are summarized in the results section for all exchange-correlation functionals discussed previously. Furthermore, bulk formation energies for each compound were calculated as follows:

$$\Delta E_{\text{form}}^{\text{AX}} = E_{\text{bulk}}^{\text{AX}} - \mu_{\text{A}} - \mu_{\text{X}} \quad (1)$$

where $\Delta E_{\text{form}}^{\text{AX}}$ is the bulk formation energy of AX-alkali halide, and μ_{A} and μ_{X} are the chemical potentials of the alkali metal and halide ion, respectively. The chemical potentials were referenced to elemental states in their respective experimentally observed lowest-energy structures (space group symbols: Li— $R\bar{3}m$, Na— $P6_3/mmc$, K— $Pnma$, Rb— $I4/mmm$, F₂— $C12/c1$, Cl₂, Br, I— $Cmce$).

For surface energy calculations, we examined two stoichiometric surface facets: (001) and (110). A symmetric slab was used for all surface models, with a 10 Å vacuum spacing added in the z -direction to prevent self-interaction between periodic images. The slab orientations and atomic arrangements are illustrated in Figure 1. Surface energies were calculated using the standard expression:

$$\sigma = \lim_{N \rightarrow \infty} \frac{1}{2A} (E_{\text{slab}}^N - NE_{\text{bulk}}) \quad (2)$$

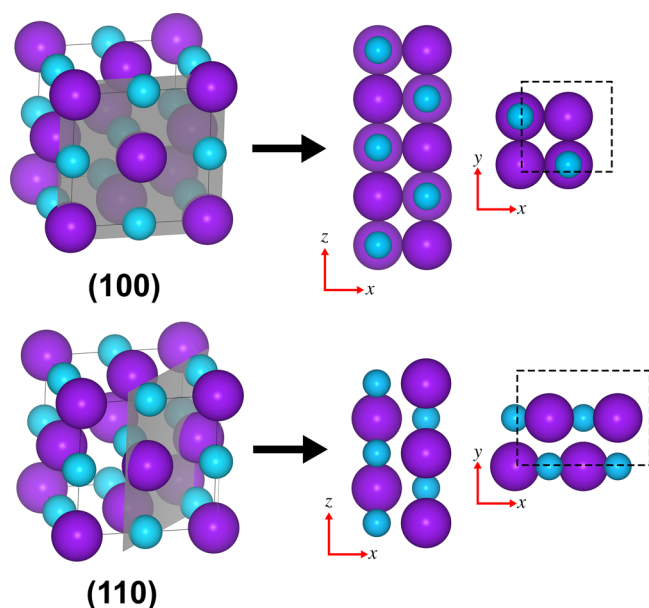


Figure 1. Surface terminations of the alkali halide (AX) crystal structure along the (100) and (110) facets, derived from the bulk structure. Purple spheres represent the cations, while cyan spheres denote the anions. Side and top views of the surface slabs are shown.

where E_{slab}^N the total energy of an N -layer slab, E_{bulk} the bulk total energy, and A is the surface area. The factor of $\frac{1}{2}$ accounts for the two surfaces of the symmetric slab. In practice, the limit is approximated using finite slab thicknesses. To avoid potential issues, such as surface energy divergence due to inconsistencies between bulk and surface energy references stemming from separate DFT calculations,^{60,61} we adopted the method proposed by Fiorentini and Methfessel.⁶² In this approach, the bulk energy (E_{bulk}) is estimated from a linear fit of the slab total energies as a function of N . Accordingly, for each AX-functional pair, slabs with 3-, 5-, 7-, and 9-layers were constructed to calculate the surface energies accurately.

3. RESULTS & DISCUSSION

3.1. Formation Energy. Understanding and accurately predicting the formation energies of solids is essential for assessing their thermodynamic stability and guiding materials design. The formation energy of a bulk material is defined as the energy change associated with forming a compound from its constituent elements in their reference states. It serves as a direct descriptor of compound stability. A more accurate measure of compound stability is decomposition energy, by which a compound's formation energy is compared to formation energies of compounds comprised of similar elements, either in different ratios or varying geometries. However, due to the high ionicity of alkali halides, there are no other stable materials comprised of the same two elements. Therefore, formation energies are a suitable metric to measure their stabilities. As alkali halides are often used in experimental calibrations and benchmarking, they serve as an ideal data set for evaluating the performance of various exchange-correlation functionals within DFT.

Table 1 presents the calculated formation energies ΔE_f of the 16 alkali halides covered in this study using five different DFT functionals: LDA, PBE, RPBE, PBE-sol, and BEEF-vdW. Figure 2 offers a visual comparison of the computed values against experimental formation enthalpies ΔH_f ,⁶³ highlighting the spread in accuracy across functionals. Among the tested functionals, BEEF-vdW consistently showed the best agreement with experimental values, yielding the lowest mean absolute error (MAE) of 35.2 kJ mol⁻¹, whereas PBE-sol was the worst-performing among the five functionals with a larger MAE of 54.2 kJ mol⁻¹. This suggests that the inclusion of van der Waals corrections, as implemented in BEEF-vdW, is beneficial for capturing subtle cohesive interactions within the alkali halide ionic crystals.

It is important to distinguish between the computed formation energies ΔE_f and the experimental formation enthalpies ΔH_f . While DFT directly yields internal energies at 0 K, experimental enthalpies account for additional thermal contributions, including zero-point energies (ZPE) and heat capacities at standard temperature and pressure. Consequently, deviations between ΔE_f and ΔH_f are expected, and so the reported MAEs might be slightly misleading. In inorganic

Table 1. Formation Energies (kJ mol⁻¹) Calculated Using LDA, PBE, RPBE, PBE-sol, and BEEF-vdW Compared to Experimental⁶³ Enthalpies of Formation

material	LDA	PBE	RPBE	PBE-sol	BEEF-vdW	experiment [63]
LiF	-603.6	-569.4	-551.5	-567.7	-581.0	-616.9
LiCl	-369.0	-354.5	-345.1	-348.7	-362.5	-408.3
LiBr	-315.8	-299.5	-292.3	-300.3	-308.1	-350.9
LiI	-243.1	-234.5	-229.5	-229.3	-238.4	-270.1
NaF	-547.1	-521.7	-510.2	-515.2	-540.4	-575.4
NaCl	-354.7	-350.8	-348.5	-343.7	-365.1	-411.1
NaBr	-309.6	-304.0	-303.6	-298.1	-316.5	-361.4
NaI	-244.2	-246.9	-249.2	-242.4	-255.2	-287.9
KF	-552.8	-524.5	-511.9	-521.1	-544.4	-568.6
KCl	-387.1	-382.0	-380.2	-374.8	-398.8	-436.7
KBr	-348.0	-341.6	-342.5	-334.5	-354.7	-393.8
KI	-288.2	-291.3	-296.0	-281.3	-304.8	-327.9
RbF	-541.7	-513.5	-500.9	-511.8	-535.0	-557.5
RbCl	-384.1	-380.4	-379.2	-373.4	-396.1	-435.4
RbBr	-347.5	-342.7	-344.2	-335.4	-356.4	-394.6
RbI	-290.4	-295.2	-301.2	-284.9	-309.4	-333.8

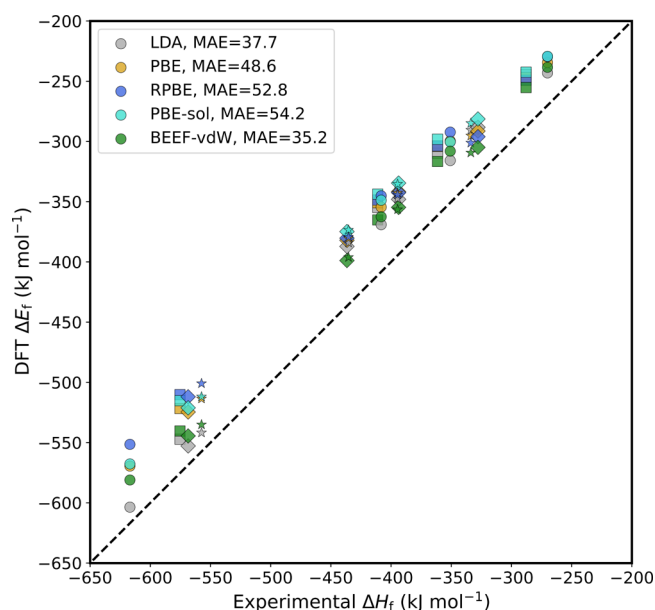


Figure 2. Calculated formation energies (kJ mol^{-1}) of 16 alkali halides versus experimental formation enthalpies (kJ mol^{-1}) using five exchange-correlation functionals: LDA, PBE, RPBE, PBE-sol, and BEEF-vdW. Li data points are shown as circles, Na as squares, K as diamonds, and Rb as stars. The mean absolute errors between DFT ΔE_f and experimental ΔH_f are reported in the legend in kJ mol^{-1} . The 45° line represents perfect agreement between experimental formation enthalpies and calculated formation energies.

systems, the deviation between the two properties is expected to be around $10\text{--}20 \text{ kJ mol}^{-1}$.⁶⁴ Despite these intrinsic differences, DFT formation energies remain valuable for relative comparisons across materials, especially when evaluating trends within chemically similar families of materials. For example, the increasing magnitude of ΔE_f from iodides to fluorides within each cation group mirrors the increasing ionic character and lattice energy. A similar but less pronounced trend is observed where ΔE_f increases as the cation size decreases for each anion group.

A notable anomaly is observed in the LDA results for alkali iodides. Specifically, while LiI shows the expected overbinding tendency,^{65,52} the heavier alkali iodides (NaI, KI, RbI) deviate from this trend, where LDA underestimates their formation energies relative to GGA functionals. This underbinding trend is caused by LDA's failure to capture the high polarizability of the large iodide anion. However, the strong polarizing power of the small lithium cation in LiI counteracts this deficiency, restoring the expected overbinding behavior.

The superior performance of BEEF-vdW is notable, particularly because this functional was originally developed with a focus on surface energies and adsorption phenomena.⁴¹ Its success in predicting formation energies suggests broader applicability to bulk properties where van der Waals interactions are non-negligible, albeit small. On the other hand, other functionals tend to underestimate formation energies of the ionic solids in this study by at least 10 kJ mol^{-1} relative to BEEF-vdW.

3.2. Lattice Constants. Lattice constants are fundamental structural parameters that directly reflect the equilibrium geometry of crystalline materials. Comparing lattice constants is important not only as a check on structural accuracy, but also because it serves as a precursor for more complex calculations.

For example, an accurate lattice constant ensures correct sampling of the Brillouin zone, better representation of electron–phonon interactions, and reduced systematic errors in adsorption energies and defect formation energies. Given the sensitivity of these properties to small changes in bond lengths, a functional's ability to reproduce experimental lattice parameters is a critical benchmark of its reliability.

Table 2 reports the computed lattice constants (in Å) for the 16 alkali halides studied here, using the same five exchange-

Table 2. Cubic Lattice Parameters (Å) Calculated Using LDA, PBE, RPBE, PBE-sol, and BEEF-vdW Compared to Experimentally Obtained Values

material	LDA	PBE	RPBE	PBE-sol	BEEF-vdW	experiment [ref.]
LiF	3.947	4.038	4.129	4.038	4.129	4.020 [66]
LiCl	5.028	5.141	5.254	5.028	5.141	5.143 [66]
LiBr	5.385	5.506	5.627	5.385	5.506	5.489 [66]
LiI	5.769	6.035	6.167	5.902	6.035	6.000 [66]
NaF	4.521	4.724	4.826	4.623	4.724	4.634 [63]
NaCl	5.526	5.774	5.899	5.650	5.774	5.640 [63]
NaBr	5.726	6.121	6.252	5.989	5.989	5.962 [66]
NaI	6.223	6.509	6.652	6.366	6.509	6.462 [66]
KF	5.132	5.368	5.604	5.368	5.486	5.360 [66]
KCl	6.074	6.354	6.633	6.214	6.354	6.293 [63]
KBr	6.369	6.662	6.955	6.516	6.809	6.586 [66]
KI	6.849	7.164	7.478	7.006	7.321	7.052 [66]
RbF	5.445	5.695	5.945	5.570	5.820	5.630 [67]
RbCl	6.397	6.691	6.985	6.544	6.691	6.570 [67]
RbBr	6.682	6.989	7.296	6.835	7.142	6.868 [67]
RbI	7.136	7.464	7.792	7.300	7.326	7.326 [67]

correlation functionals. These are compared against experimental values and visualized in Figure 3. Across all systems,

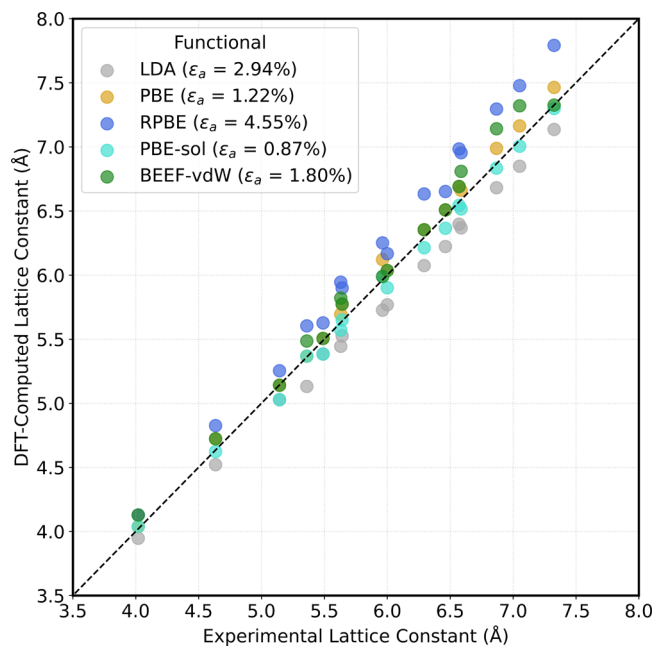


Figure 3. Calculated lattice constants (Å) of the cubic unit cell of alkali halides using five exchange-correlation functionals: LDA, PBE, RPBE, PBE-sol, and BEEF-vdW. The 45° line represents perfect agreement between experiment and DFT. The average percent error is reported in the legend for each functional.

general trends are well-captured by all functionals. For example, the lattice constants increase as the anion size increases from F[−] to I[−]. However, the magnitude of agreement with experiment varies significantly between functionals. Among the functionals tested, PBE-sol exhibits the closest agreement with experiment across all 16 alkali halides tested, with a MAE of 0.052 Å and a percent error of 0.87%. This is consistent with its design purpose: to improve solid-state predictions by correcting for the systematic overestimations of standard PBE in dense materials.⁴⁰ Its performance is robust across both light and heavy halides, unlike other functionals, which perform much worse with heavier halides.

PBE is slightly less accurate than PBE-sol, but still performs reliably well with an MAE of 0.075 Å and a percent error of 1.22%. It is a reasonable choice for calculating lattice constants, especially due to its simplicity and availability in many DFT codes. In contrast, RPBE exhibits the worst performance of all five functionals studied in this work. With a substantial MAE of 0.279 Å and a percent error of 4.55%, it systematically overestimates the lattice constants. This trend is consistent with RPBE's design, which reduces the strength of the exchange interaction to prevent the overbinding observed in standard PBE and more specifically on adsorption systems.³⁹ However, for bulk ionic crystals like alkali halides, this under-binding leads to weakened bonding, and consequently resulting in artificially inflated unit cells.^{68–70} As such, RPBE should be avoided for such systems. In comparison, LDA demonstrates its characteristic overbinding tendency,^{71,72,73} yielding the second-highest error (MAE $\approx$ 0.177 Å, $\epsilon_a \approx$ 2.94%). It performs much better for smaller systems, such as LiF and NaF. However, its deviations become more pronounced for heavier halides like RbBr and RbI.

BEEF-vdW, which performed exceptionally well in reproducing experimental formation energies, tends to modestly overestimate lattice parameters. While its performance for mid-sized halides (i.e., NaCl, KBr) is reasonably close to experiment, its overall accuracy is less consistent than PBE-sol, with slightly larger errors (MAE $\approx$ 0.108 Å, $\epsilon_a \approx$ 1.80%).

3.3. Surface Energy. Using the methods described above, surface energies (in erg cm^{−2}) for the (100) and (110) facets of all 16 alkali halides were calculated using the five different exchange-correlation functionals. The results are tabulated in Tables 3 and 4, and visually presented in Figure 4. For qualitative reference, Tasker's classical model values are included as black dashed lines. These classical predictions should not be treated as strict benchmarks, as they rely on simplified ionic models rooted in classical electrostatics. Nevertheless, their close agreement with several of the DFT-based results is striking, given the model's simplicity and its historical role in describing the energetics of alkali halide surfaces.

Systematic trends across the functionals are readily apparent and consistent across both facets. LDA uniformly overestimates surface energies relative to Tasker's computed values, predicting the largest values in every single case across the data set. This can be explained by LDA's tendency to overbind due to its lack of gradient corrections, which leads to stronger surface cohesion and larger surface energies. On the other hand, RPBE consistently underestimates surface energies relative to Tasker's values, producing the lowest values across the entire data set, stemming from RPBE's contrasting under-binding behavior. The remaining three functionals occupy a middle ground. With a few exceptions, the calculated surface energies of all alkali halides follow a consistent trend in surface energy predictions: LDA > BEEF-vdW > PBE-sol > PBE > RPBE. LDA consistently yields

Table 3. Calculated Surface Energies (erg cm^{−2}) of the (100) Facet of Alkali Halides Using Five Exchange-Correlation Functionals: LDA, PBE, RPBE, PBE-sol, and BEEF-vdW^a

material	LDA	PBE	RPBE	PBE-sol	BEEF-vdW	Tasker
LiF	498.7	279.6	192.7	367.8	378.9	416
LiCl	268.3	126.1	67.7	169.1	195.2	177
LiBr	231.7	102.5	57.2	144.0	166.4	137
LiI	181.8	82.7	31.9	130.1	144.6	102
NaF	403.2	268.7	219.2	284.6	339.9	321
NaCl	245.3	153.9	116.5	172.2	277.7	180
NaBr	199.0	126.6	90.9	146.3	175.7	151
NaI	168.9	88.7	54.5	115.8	146.1	105
KF	268.9	169.9	151.5	202.4	256.9	230
KCl	188.2	117.9	101.3	129.8	183.5	162
KBr	167.9	103.3	87.0	111.3	165.5	142
KI	143.7	83.0	67.4	97.1	142.0	122
RbF	233.4	140.6	121.9	158.7	215.7	188
RbCl	166.9	104.9	88.9	115.8	159.2	148
RbBr	149.9	91.8	77.7	94.5	148.6	132
RbI	129.3	75.5	62.2	87.1	129.4	115

^aFor comparison, Tasker's classical model surface energies are included in the final column.

Table 4. Calculated Surface Energies (erg cm^{−2}) of the (110) Facet of Alkali Halides Using Five Exchange-Correlation Functionals: LDA, PBE, RPBE, PBE-sol, and BEEF-vdW^a

material	LDA	PBE	RPBE	PBE-sol	BEEF-vdW	Tasker
LiF	1078.1	820.2	701.8	902.9	873.4	975
LiCl	533.5	372.8	300.7	432.0	439.4	420
LiBr	442.4	301.4	235.5	360.5	361.1	326
LiI	347.0	225.4	166.3	282.9	284.4	255
NaF	825.2	632.2	558.4	671.6	707.7	717
NaCl	461.1	340.4	291.9	370.6	408.1	392
NaBr	389.7	278.6	232.7	310.5	339.2	325
NaI	309.2	211.5	168.4	244.2	267.9	231
KF	566.7	420.3	367.1	449.9	490.0	471
KCl	365.1	266.3	230.5	286.9	337.1	321
KBr	318.2	226.9	195.0	250.2	290.0	278
KI	261.3	181.9	152.6	202.1	239.5	241
RbF	482.6	348.2	300.7	378.9	411.0	401
RbCl	322.3	233.3	200.8	252.6	293.4	288
RbBr	284.4	202.9	173.7	221.2	258.4	253
RbI	237.2	164.4	139.6	182.5	218.3	215

^aFor comparison, Tasker's classical model surface energies are included in the final column.

the highest surface energies, followed by BEEF-vdW, PBE-sol, and PBE, while RPBE predicts the lowest surface energy values for a given compound.

The consistent positioning of BEEF-vdW and PBE-sol to Tasker's classical surface energies across most halides suggests that these functionals strike a reasonable balance between accuracy and transferability. While Tasker's values are not derived from first-principles, the qualitative agreement indicates that these functionals reproduce classical trends more closely than others. This close agreement is also visually evident in Figure 4, where the calculated values from these two functionals closely trace Tasker's estimates, with mean absolute errors of 24.4 (BEEF-vdW) and 25.9 (PBE-sol) erg cm^{−2}. However, it is important to note that classical models may not serve as definitive benchmarks for surface energies, and thus, while

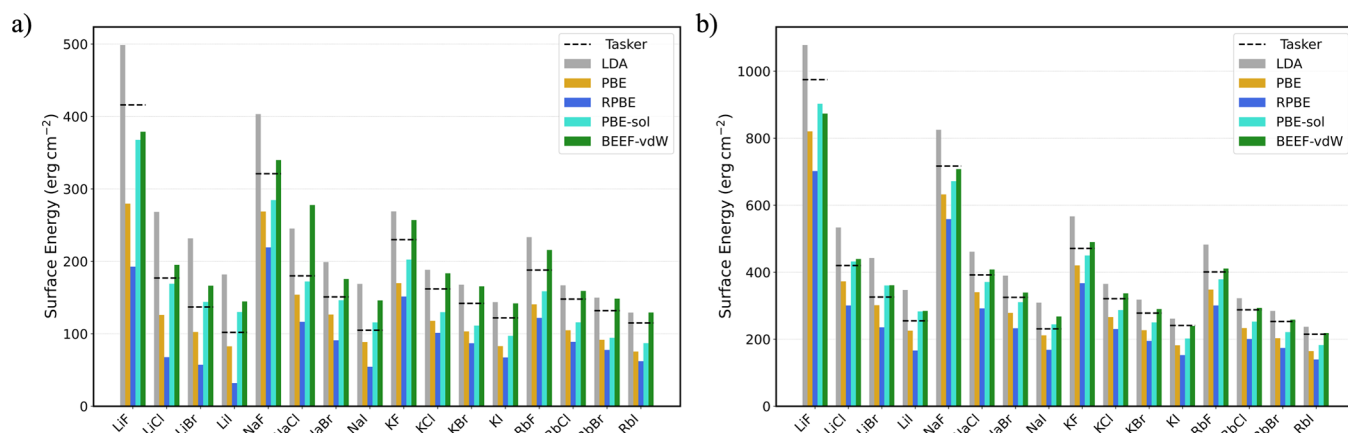


Figure 4. Calculated surface energies (erg cm^{-2}) of alkali halides using five exchange-correlation functionals: LDA, PBE, RPBE, PBE-sol, and BEEF-vdW. Panels (a) and (b) present the calculated values for the (100) and (110) facets, respectively.

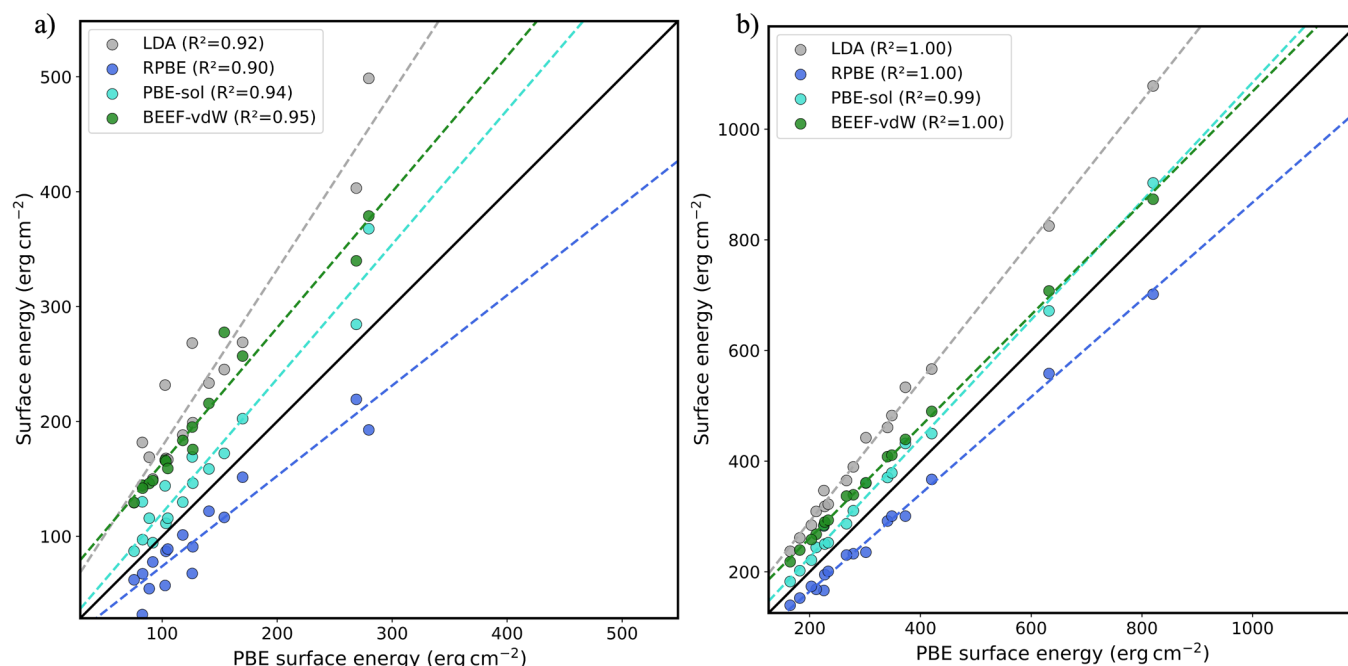


Figure 5. Correlations between surface energies calculated using LDA, RPBE, PBE-sol, and BEEF-vdW exchange-correlation functionals and PBE for (a) the (100) facet and (b) the (110) facet of alkali halides.

agreement is encouraging, it should not be interpreted as a conclusive indicator of higher accuracy.

Importantly, these findings highlight the robustness of functional-dependent behavior across diverse properties. The same general ordering observed in bulk lattice predictions is also observed in surface energy calculations. This concludes that the tendencies of each functional toward under- or overbinding manifest consistently in both bulk and surface regimes. This consistency lends confidence to using a single functional across multiple stages of a simulation workflow, particularly when comparing relative trends across materials. However, the overwhelming use case of surface calculations is to study adsorption phenomena, which are rather difficult to benchmark due to experimental complexity and are excluded from this study.

Figure 5 presents a comparison of surface energies calculated using LDA, RPBE, PBE-sol, and BEEF-vdW relative to PBE, for both the (100) (panel a) and (110) (panel b) facets. PBE is used

as a reference because it is one of the most widely adopted GGA functionals in DFT and serves as a standard baseline for evaluating the performance of other functionals. In each panel, the 1:1 diagonal black line represents perfect agreement with PBE, while colored dashed lines correspond to linear regression fits for each functional. The R^2 values provide a measure of how closely the functionals track the PBE values across all 16 alkali halides. The figure reveals a striking contrast between the two facets in terms of the consistency of functionals. In panel b, the (110) facet surface energies calculated by all functionals exhibit extremely high correlation with PBE, with R^2 values approaching or equaling 1.00. This perfect alignment suggests that the relative scaling of surface energies is robust. The absolute values may differ, but the rank ordering of surface energies across materials is preserved with all the functionals.

In contrast, panel a shows more variation in correlation behavior for the (100) facet, where R^2 values span a wider range of 0.90 to 0.95. LDA and RPBE show notably weaker correlation

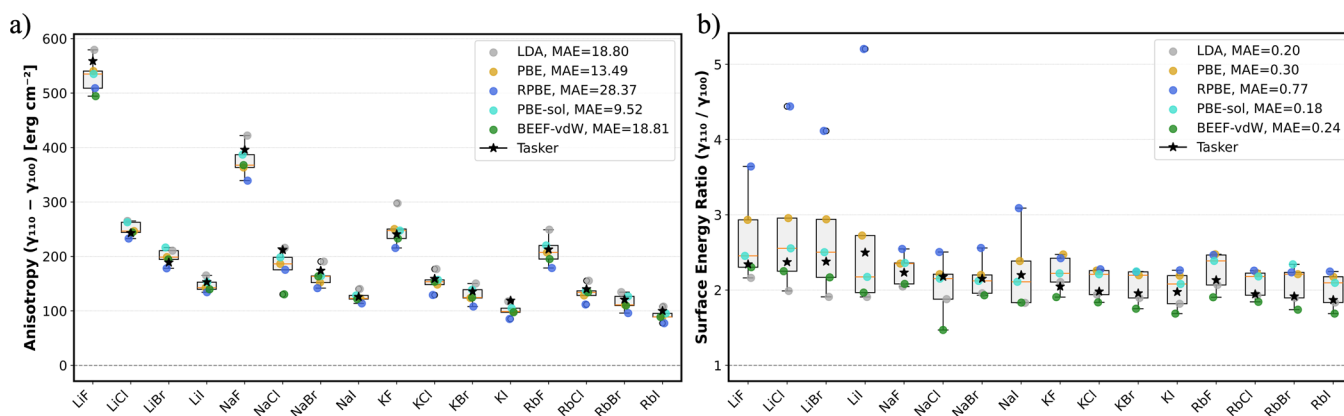


Figure 6. (a) Anisotropy values (erg cm^{-2}) and (b) (110):(100) surface energy ratios for alkali halides. Both values are calculated using five exchange-correlation functionals: LDA, PBE, RPBE, PBE-sol, and BEEF-vdW.

to PBE than PBE-sol and BEEF-vdW. This dispersion indicates that the (100) facet is more sensitive to the exchange-correlation functional than the (110) facet, leading to nonuniform shifts in surface energies. This is particularly evident for LDA and RPBE, which tend to overbind and under-bind. The contrast in the behavior of LDA and RPBE on either surface can be explained by the surface physics of each surface.

The (100) facets in alkali halides are more compact and symmetric, which leads to subtler surface relaxations and more pronounced sensitivity to the treatment of short-range exchange-correlation effects and gradient corrections. As a result, functionals like LDA and RPBE, which are known to either overbind or overcorrect gradient effects, tend to deviate significantly from PBE-sol and BEEF-vdW on these surfaces. In contrast, the (110) facet undergoes larger relaxations, as evidenced by Figure S1, which may lead to a surface energy that is less dependent on the choice of functional. Consequently, LDA and RPBE show relatively better agreement with more sophisticated functionals for the (110) surface energies. These assumptions are only valid if the surfaces remain in their pristine terminated form, without undergoing any reconstruction or adsorption-induced modifications. Following the benchmark against PBE in Figure 5, an analogous analysis is presented in Figure S3 using PBE-sol as the reference. Plotting against PBE-sol more effectively reveals the deficiencies of certain methods, highlighting the severe errors introduced by an inappropriate functional like RPBE when calculating alkali halide surface energies.

To summarize, the functional behavior observed for bulk lattice constants carries over to surface energetics: LDA overbinds, RPBE under-binds, and PBE, PBE-sol, and BEEF-vdW fall in between, with the latter two consistently offering best agreement with experiment. Accurately evaluating surface energies is crucial in systems where relative accuracies of different facets are necessary, such as electrochemical interfaces, crystal growth, and nanoparticle modeling. To assess the practical implications of functional choice beyond numerical accuracy, we quantify two metrics aimed at evaluating whether the selected exchange-correlation functional can influence qualitative scientific judgments. These include decisions commonly made in surface science and materials modeling, such as identifying the most stable surface facet of a material. While quantitative agreement with experiment is important, such qualitative trends often guide structural modeling and catalyst design assumptions in simulations.

Figure 6 provides two perspectives on quantifying surface energy differences between facets. Panel a presents the absolute anisotropy, defined as the difference in surface energy between the (110) and (100) facets of a given material:

$$\Delta\sigma_i = \sigma_{i,(110)} - \sigma_{i,(100)} \quad (3)$$

whereas panel b presents a surface energy ratio:

$$R_i = \frac{\sigma_{i,(110)}}{\sigma_{i,(100)}} \quad (4)$$

Figure 6 presents the data for both metrics, calculated using the same five exchange-correlation functionals for all 16 alkali halides in this study, where i denotes a specific alkali halide compound.

Figure 6a shows that the anisotropy is largest for lighter halides, such as LiF, LiCl, and NaF, where the differences between (110) and (100) surface energies exceeds 200 erg cm^{-2} and, in the case of the lightest halide, exceeds 500 erg cm^{-2} . As we move down the periodic table toward heavier halides, the anisotropy steadily decreases, indicating that the energy differences between different facets narrow. This behavior likely stems from the softer ionicity of larger ions making up the alkali halide, which increases its polarizability and thus reduces the directional dependency of surface stability.

Despite the large differences in surface energies predicted by each functional, all functionals preserve the qualitative anisotropy trend across all 16 alkali halides. They are tightly packed, which implies they all exhibit similar qualitative trends on describing surface stability. Among the functionals, PBE-sol and BEEF-vdW again yield the most similarity relative to Tasker's classically calculated values, underlining their consistency.

Figure 6b shows the surface energy ratio between the (110) and (100) facets for each alkali halide. This dimensionless metric provides a normalized measure of surface stability, where a value greater than 1 indicates that the (100) facet is more stable than the (110) facet, which is generally expected for cubic ionic crystals. Consistent with this expectation, all alkali halides strongly prefer the (100) facet with most ratios falling in the range of 1.8–3. A few systems exhibit even higher ratios, exceeding 5 in rare cases.

RPBE stands out among all functionals for producing significant outliers, particularly for lighter halides such as LiI and LiBr. The discrepancy is even more pronounced on LiI, where a small cation (Li^+) is paired with a larger, more

polarizable anion (I^-). This asymmetry in ion size and polarizability magnified the under-binding tendencies of RPBE. The (110) facet becomes excessively destabilized due to its lower coordination, thereby distorting the relative stability between facets. Interestingly, RPBE performs more reasonably at the opposite end of the spectrum, such as in RbF, where a larger cation (Rb^+) is paired with a small, hard anion (F^-). In this case, the electrostatic interactions are much stronger, reducing the sensitivity of surface energetics to the functional's exchange treatment, and resulting in a more balanced prediction of the (110):(100) surface energy ratio.

These results highlight the importance of selecting an appropriate exchange-correlation functional when modeling surface energetics. In many cases, inaccuracies in predicting the absolute surface energies of different facets may partially cancel, still allowing for a qualitatively correct conclusion about facet stability. However, when a functional introduces asymmetric or disproportionate errors, as demonstrated with RPBE, this error cancellation breaks down, potentially leading to qualitatively incorrect predictions of facet preference. Such inconsistencies can lead to significantly varying interpretations of Wulff constructions and surface reactivity. Therefore, ensuring that the chosen functional yields a consistent performance across different material types is critical to the physical reliability of DFT-based models.

4. CONCLUSIONS & OUTLOOK

In this work, we conducted a comprehensive benchmark of five exchange-correlation functionals (LDA, PBE, RPBE, PBE-sol, BEEF-vdW) in predicting the bulk and surface properties of 16 alkali halides. By systematically analyzing lattice constants, enthalpies of formation, and surface energies of the low-index (100) and (110) facets, we assessed not only the numerical accuracy of each functional, but also their internal consistency and robustness across chemical space and whether they impact the qualitative conclusions relevant to surface science and materials modeling.

Our results show that PBE-sol and BEEF-vdW consistently offer the best balance of accuracy and reliability for both bulk and surface predictions, providing the lowest mean absolute errors for lattice constants and surface energies. A notable limitation of PBE-sol is its tendency to underestimate formation energies. This trend was also observed in other functionals, though to varying extents. LDA, while known for overbinding, demonstrated high accuracy for lighter halides, but systematically overestimated surface energies across the board relative to other GGA functionals and Tasker's classical estimates. By contrast, RPBE significantly under-binds relative to other GGA functionals, LDA, and Tasker's estimates. It introduced significant deviations, particularly in systems with asymmetric ion sizes. LDA and RPBE had much larger errors than the other three functionals in predicting bulk lattice constants. PBE, while less accurate than PBE-sol in absolute terms, remained consistent and usable for all qualitative assessments.

In conclusion, this work provides a practical assessment of commonly used DFT functionals for modeling alkali halides. The insights presented can inform the selection of functionals for simulations involving ionic materials, surface reconstructions, crystal growth, and morphology predictions. The observed trends highlight the ongoing need for functional benchmarking in systems and ensure that a chosen functional can consistently and accurately describe a system's behavior.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsomega.5c06914>.

Computational parameters; convergence tests; and relative displacements (PDF)

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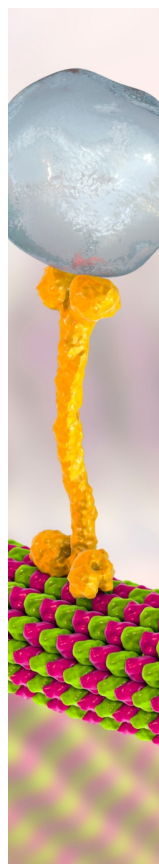
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